

Crash Busters

Sherrine Fils-Aime, Sarah Morais, Yemima Dort, Shoan Andrew Alex Lnu,

Mihir Bagadia, Trent Burger, Ian Cueli

ISM 6423: Data Analysis for Decision Support

Professor Liangfei Qiu

April 20, 2026

Introduction

Gainesville is a unique city compared to many others in the state of Florida, mainly due to it being the home of the University of Florida which directly and indirectly affects the entire city's operations, but it brings a unique microcosm of individuals, people, and issues to address within city limits. The city ebbs and flows due to much of the population constantly changing year after year (students coming to the city, leaving on breaks, graduating and leaving or staying, families and young adults matriculating to and from, etc.,) which also means that the trends in everyday life differ as well.

One of the biggest areas that Gainesville differs across all these different population changes is traffic. Traffic is one of the biggest areas for concern and an aspect of daily life that everyone living here must interact with daily, with some of the biggest traffic concerns coming from areas that are closest to the heart of UF. Traffic in Gainesville, unlike most other cities in Florida that are very car-centric, involves a wider range of road users such as cyclists, pedestrians, vehicles, mopeds, buses, etc., all of which make traffic changes much more difficult to navigate.

Population Growth & Driving Factors

Gainesville, Florida, continues to grow at a steady rate at around 1.07% each year, according to the U.S. Census Bureau.¹ With a current population of about 151k residents, the population increased by about 6.8% since the most recent census in 2020, recording a population of about 142k residents.² Gainesville is projected to continue to grow at a similar rate over the next 10 years, with key drivers including the expansion of the University of Florida, UF Health, and new developments, especially in west Gainesville near Interstate 75 around Newberry Road and Archer Road.

Additionally, excluded from the Gainesville population, are the metropolitan areas surrounding Gainesville where many residents commute to Gainesville for work or resources, such as UF Health or Butler Plaza. This includes the areas in Alachua County and Gilchrist County, with cities such as Newberry, Archer, Alachua, High Springs, Trenton, etc. With over 32,000 people employed at the University of Florida, a significant portion of staff commute from outside of Gainesville.^{3,4}

According to a survey from the U.S. Census Bureau in 2024, driving to work alone is the most popular form of commuter transportation with about 60.9% of residents, followed by 14.7% of residents who work from home, and 9.2% of residents who carpool.⁵ Between 2018 and 2024, residents who worked from home increased about 11%, with about 3% increasing before COVID-19 and about 8% increase after COVID-19. It's important to note that this survey was completed in 2024, another shift in policy may impact the number of commuters and traffic in Gainesville when UF mandated a return to in-person work in July 2025, impacting about 3,500

employees who were working either fully remote or in a hybrid capacity since the COVID-19 shutdown in 2020.⁶

When looking at trends for commuting with other modes of transportation from 2013-2024, public transit and motorcycle usage decreased in popularity, while mopeds and other forms of transportation increased. Bicycles have seen a decrease in popularity until 2020, when popularity started to increase again, along with walking as a mode of transportation.

Electronic scooters and electronic bikes started to surge in Gainesville in 2021 when the city launched a shared mobility program in partnership with UF.⁷ Rental options through companies such as Veo, Bird and Spin started popping up around Gainesville, especially near campus and downtown locations. Within the first year, nearly 200,000 trips were taken with the adoptable scooters and bikes. There was also a surge simultaneously with the purchase of e-scooters and e-bikes for students and commuters. This led to Gainesville investing in infrastructure for dedicated bike lanes and a growing network of charging stations to better support micromobility solutions.

Traffic safety and infrastructure development is a key priority for the city of Gainesville. Over the past 10 years, Gainesville has invested in wider sidewalks, additional crosswalks, eliminating parking on the street, curbside pull-out stops for buses, additional bike lanes, changing roads to one-way roads, adding roundabouts, speed bumps, reducing speed limits, ADA improvements for bus stops, red-light cameras, etc.⁸ In 2022, Alachua County residents voted to increase sales tax by one-cent, where a large portion of the surtax proceeds are dedicated to road repairs and public-safety buildings.⁹ Additionally, in 2018, the City Commission adopted a commitment to safety with the Vision Zero plan, seeking to eliminate traffic-related deaths and reduce serious injuries by 2040. Since its adoption, the city has invested over \$3 million in its effort, hiring dedicated staff and adopting policies to strengthen its commitment.¹⁰

Key Questions

This research project focuses on a few key issues to help policy makers in the city of Gainesville in decision making around traffic safety. For example, to what extent does the frequency and severity of crashes of motor vehicles differ from the frequency and severity of crashes involving vulnerable users (pedestrians and cyclists) in Gainesville? From the analysis associated with this research question and based on the crashes of vulnerable users, we would then also like to expand our analysis by exploring what are the primary geographic "hotspots" and peak temporal windows for traffic accidents in Gainesville, and how can these identify the highest-priority corridors for city planning?

Methodology/Analysis

This secondary data analysis uses data collected from Gainesville's Open Data Portal, titled 'Traffic Crashes', which has catalogued information on every traffic crash beginning in 2011 that occurred within the City of Gainesville limits and was within the jurisdiction of the Gainesville Police Department. For the purposes of this analysis, the data we will be utilizing spans from 2015 to 2025, with a comment on trends for the few months of data present for 2026. The dataset has been added to and maintained by GPD, with the relevant data points and columns assumedly derived from their internal reporting systems after a crash.

Data Cleaning, Preparation, and Transformation

To prepare the dataset for exploratory data analysis, chi-squared, and regression analysis, we performed data cleaning and standardizing for several variables. First, we transformed the Crash Date column, which is the date of the crash as determined by the reporting officer, from a floating timestamp into a date time variable, making it more usable format for plotting. With this transformation, time of day bins or sections throughout the day were defined (Morning, Afternoon, Evening, and Night). We then subsetted the dataset to pull the relevant rows that we needed for our time span on 2015 to 2025, which corresponded to 58,032 crashes and 390 crashes so far in 2026. Along with subsetting for these rows, it was necessary to compare relevant data points over the years but also at the 5 and 10 year mark, so a classification for 2015-2019 and 2020-2025 was also outlined in our analysis.

Along with preparing the dataset for analysis, we also created two new columns to identify key points within our analysis. The first column is **Crash Type**, where a function in Python was defined to classify crashes first on if they included a bicycle or a pedestrian, and then they were segmented into four different crash types or categories: Multi-Vulnerable (the crash included both cyclists and pedestrians), Involving Cyclist, Involving Pedestrian, and Motor Vehicle Only. The second column we defined was **Severity**, where a function classified each row in the data frame on if the crashes were considered Fatal (Total Fatalities was ≥ 1), Probable Injury (Total People ≥ 2), and if no fatalities and less than 2 people were involved, the crash would be classified as Property Damage Only.

Notes on the Dataset

There are a few notes from our dataset that are important to discuss as they give a commentary on the pros and cons of an open source dataset, especially one from internal city systems. The first note to discuss is related to the sample size discrepancies in our analysis groups. For the three main groups we are conducting analysis into (bicycles and pedestrians vs motor vehicles, fatalities vs non-fatalities, intersections vs not at an intersection), the discrepancies in the sample

sizes are quite large (e.g., there are over 35,000 crashes not at an intersection vs less than 20,000 crashes at all other intersection types).

These unequal sample sizes suggest potential confounders not represented in our dataset and can lead to a loss in power as well as unequal variances, which can be seen in the lack of variation explained for crash type and fatality in our analysis. This doesn't mean that our results aren't significant, but that a more expansive analysis may better explain our main research question. This more expansive analysis could have also included missing variables from our dataset that may better explain these variations such as age, demographics, zip code, prior traffic history, etc.

For our geographical maps and visualizations, there are three notes that are important to represent as well to better contextualize our GIS visualizations. For one, traffic crashes and data points on UF's main campus were not included in this dataset as the University of Florida is a special entity and is subject to a different jurisdiction than the rest of Gainesville (GPD vs UFPD). Two, there are discrepancies with how the data is entered, e.g., SR 24 (SW Archer Road) vs SW Archer Road which skewed some of the results and visualizations. Finally, the geographical data recorded has a pretty big error, for crashes that the system has not been able to identify, it assigns a default coordinate (29.65198, -82.32272) which points to the Alachua County Clerk of the Court's Office, so we had to strip these rows as to not skew the maps.

Data Analysis/Results

All of our analysis and visualizations were conducted in Python, through using Google Colab and ArcGIS for the geographical visualizations.

Overview of Crash Data (2015–2025)

The analysis covers crash data from 2015 to 2025 in Gainesville, which is characterized by a significant class imbalance. The vast majority of incidents are “Motor Vehicle Only: crashes (55,562 cases). “Probable Injury” is the most frequent outcome, while “Fatal” crashes are rare (150 cases).

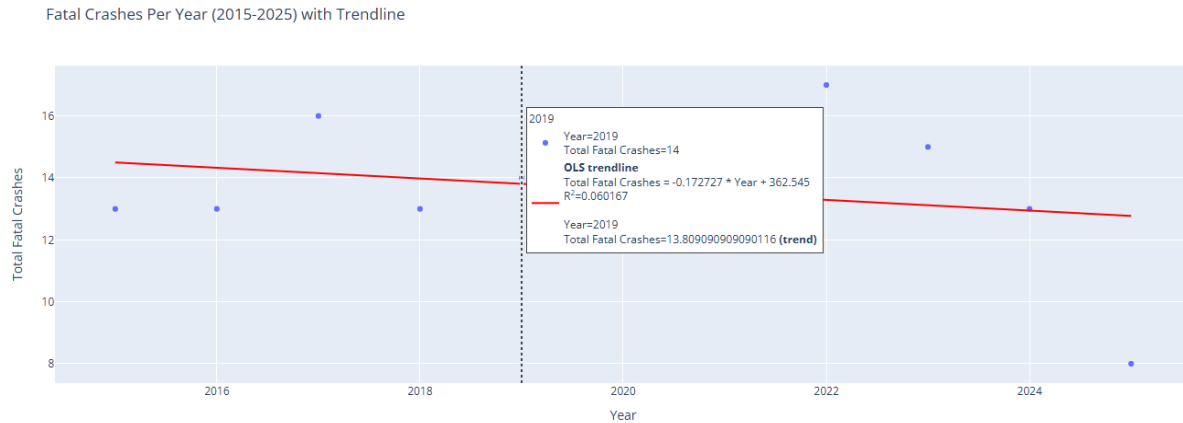
Feature engineering complete.

Crash Type distribution (2015-2025):

	Crash Type	Count
0	Motor Vehicle Only	55562
1	Involving Pedestrian	1297
2	Involving Cyclist	1171
3	Multi-Vulnerable	2

Severity distribution (2015-2025):

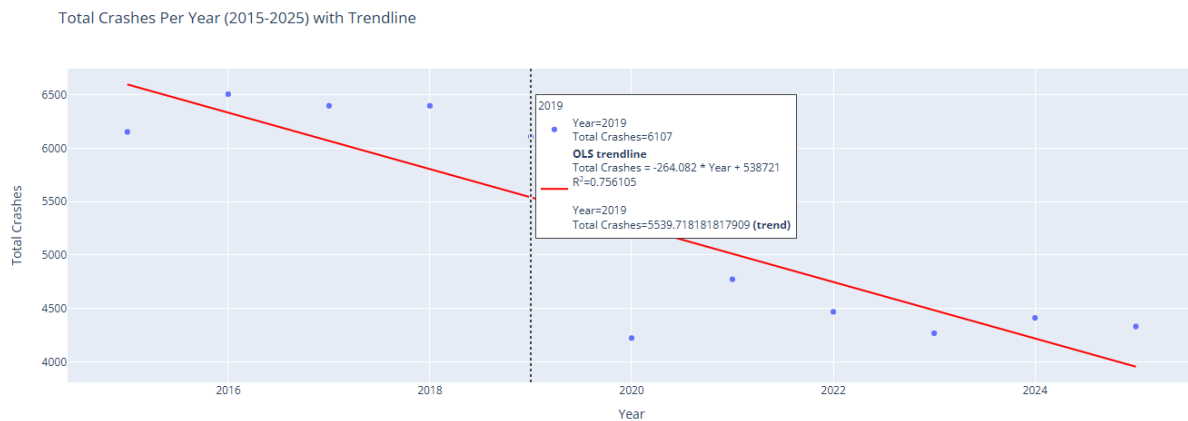
Severity	
Probable Injury	51404
Probable Property Damage	6478
Fatal	150



Trends in Crash Frequency

Total Crashes:

- **Overall Trend:** There is a clear and significant **downward trend** in the total annual number of crashes over the ten-year period.
- **Fluctuations:** While the data shows considerable year-over-year variability—including a notable dip in 2020 due to the COVID-19 pandemic—the long-term trajectory indicates a sustained reduction in traffic incidents.



Fatal Crashes:

- **Stability vs. Decline:** The frequency of fatal crashes has been relatively stable across the study period, showing a less pronounced overall decline compared to total crashes.
- **Peak/Minimum:** Increases were noted in 2017 and 2022, but a historical minimum for fatalities was documented in 2025.

Crash Type Analysis

- **Dominant Type:** The majority of the crashes are “Motor Vehicle Only.”
- **Annual Changes:** Over the years, both “Motor Vehicle Only” and “Involving Cyclist” crashes have generally decreased.
- **Counter Trend:** In contrast, “Involving Pedestrian” crashes show a slight increase in the later years of the study.

Statistical Tests Performed

Chi-square Test

- **Statistical Significance:** There is a statistically significant association between the type of crash and the likelihood of a fatality (p-value < 0.05).
- **High-Risk Types:** Crashes involving pedestrians and cyclists have a higher proportion of fatal outcomes compared to “Motor Vehicle Only” incidents.

A Chi-square test was conducted using a **contingency table** to evaluate the relationship between crash categories and fatality outcomes.

Ho: There is no statistically significant association between crash type and fatality outcomes.

Ha: There is a statistically significant association between crash type and fatality outcomes.

The analysis produced a Chi-square statistic of **609.41** and a very low p-value, which permitted the rejection of the null hypothesis. The results show a **statistically significant association** between the specific crash typology and fatality outcomes, demonstrating that the risk of mortality is not uniform across all crash types.

Contingency Table for Chi-square Test:

Severity	Fatal	Non-Fatal
Crash Type		
Involving Cyclist	11	1160
Involving Pedestrian	47	1250
Motor Vehicle Only	92	55470
Multi-Vulnerable	0	2

Chi-square Statistic: 609.41
P-value: 9.211e-132
Degrees of Freedom: 3

Logistic Regression Models

To analyze the factors contributing to fatal accidents, multiple logistic regression models were conducted using Maximum Likelihood Estimation (MLE). The models evaluate the influence of crash types and location characteristics on the probability of a fatal outcome.

Logit Regression Results						
Dep. Variable:	IsFatal	No. Observations:	58032			
Model:	Logit	Df Residuals:	58028			
Method:	MLE	Df Model:	3			
Date:	Tue, 14 Apr 2026	Pseudo R-squ.:	0.09403			
Time:	14:19:45	Log-Likelihood:	-945.40			
converged:	False	LL-Null:	-1043.5			
Covariance Type:	nonrobust	LLR p-value:	2.740e-42			
	coef	std err	z	P> z	[0.025	0.975]
const	-4.6583	0.303	-15.377	0.000	-5.252	-4.065
CrashType_Involving Pedestrian	1.3775	0.337	4.083	0.000	0.716	2.039
CrashType_Motor Vehicle Only	-1.7435	0.320	-5.442	0.000	-2.372	-1.116
CrashType_Multi-Vulnerable	-19.7851	1.44e+05	-0.000	1.000	-2.82e+05	2.82e+05

H₀: Crash type is not a statistically significant predictor of accident severity.

H_a: Crash type is a statistically significant predictor of accident severity.

This logistic regression shows crash type is a highly significant predictor of accident severity ($p < 0.05$). **Pedestrian involvement** (coefficient = 1.3775, $p < 0.001$) increases the log-odds of a fatal outcome. Conversely, **Motor Vehicle Only** crashes are significantly safer than the baseline (cyclists) with beta = -1.7435 and $p < 0.001$. The **Multi-Vulnerable** variable was statistically insignificant ($p = 1.000$) due to the lack of fatal incidents in that category.

Logit Regression Results						
Dep. Variable:	Severity	No. Observations:	58026			
Model:	Logit	Df Residuals:	58018			
Method:	MLE	Df Model:	7			
Date:	Tue, 14 Apr 2026	Pseudo R-squ.:	0.005668			
Time:	14:20:10	Log-Likelihood:	-1037.6			
converged:	False	LL-Null:	-1043.5			
Covariance Type:	nonrobust	LLR p-value:	0.1063			
	coef	std err	z	P> z	[0.025	0.975]
const	-2.9957	1.025	-2.924	0.003	-5.004	-0.987
IntersectType_FOUR-WAY INTERSECTION	-2.8313	1.037	-2.729	0.006	-4.865	-0.798
IntersectType_NOT AT INTERSECTION	-3.1028	1.031	-3.011	0.003	-5.123	-1.083
IntersectType_OTHER (EXPLAIN IN NARRATIVE)	-3.0366	1.176	-2.581	0.010	-5.342	-0.731
IntersectType_ROUNDABOUT	-3.2229	1.432	-2.250	0.024	-6.030	-0.415
IntersectType_T-INTERSECTION	-2.4656	1.047	-2.356	0.018	-4.517	-0.414
IntersectType_TRAFFIC CIRCLE	-19.7119	7234.304	-0.003	0.998	-1.42e+04	1.42e+04
IntersectType_Y-INTERSECTION	-3.2048	1.432	-2.237	0.025	-6.012	-0.397

H₀: Intersection type is not a statistically significant predictor of accident severity.

H_a: Intersection type is a statistically significant predictor of accident severity.

This analysis shows that intersection types like **Four-Ways**, **T-Intersections**, and **Roundabouts** are significantly associated ($p < 0.05$) with reduced high-severity outcomes compared to the baseline. However, the overall model is not significant because its p-value is 0.1063, so we fail to reject the null hypothesis. Therefore, intersection type is not a statistically significant predictor of accident severity.


```

=== LOGISTIC REGRESSION RESULTS ===
                        Logit Regression Results
=====
Dep. Variable:          IsFatal    No. Observations:          58030
Model:                  Logit      Df Residuals:              58026
Method:                 MLE        Df Model:                  3
Date:                   Tue, 14 Apr 2026    Pseudo R-squ.:            0.09447
Time:                   14:20:13    Log-Likelihood:           -944.94
Converged:              True        LL-Null:                  -1043.5
Covariance Type:        nonrobust    LLR p-value:              1.728e-42
=====

```

	coef	std err	z	P> z	[0.025	0.975]
const	-4.7655	0.323	-14.732	0.000	-5.400	-4.132
Type_Involving Pedestrian	1.4072	0.339	4.154	0.000	0.743	2.071
Type_Motor Vehicle Only	-1.6958	0.324	-5.232	0.000	-2.331	-1.060
AtIntersection	0.1632	0.168	0.972	0.331	-0.166	0.493

```

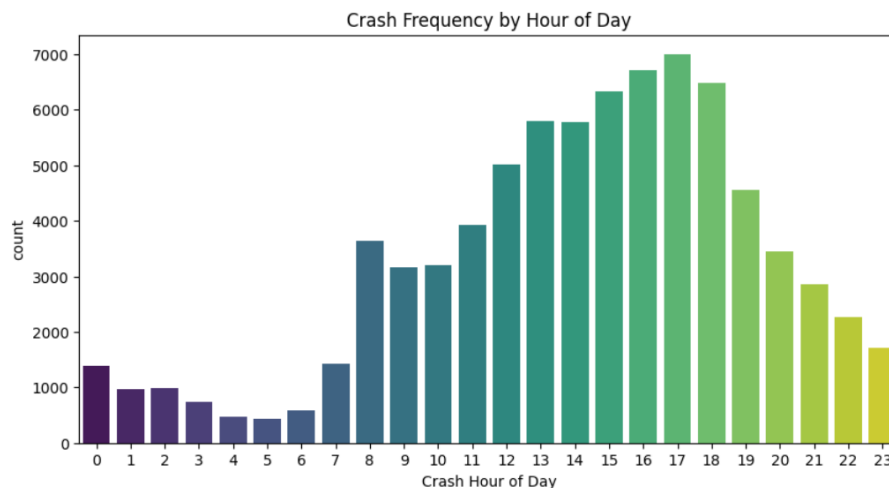
=====

```

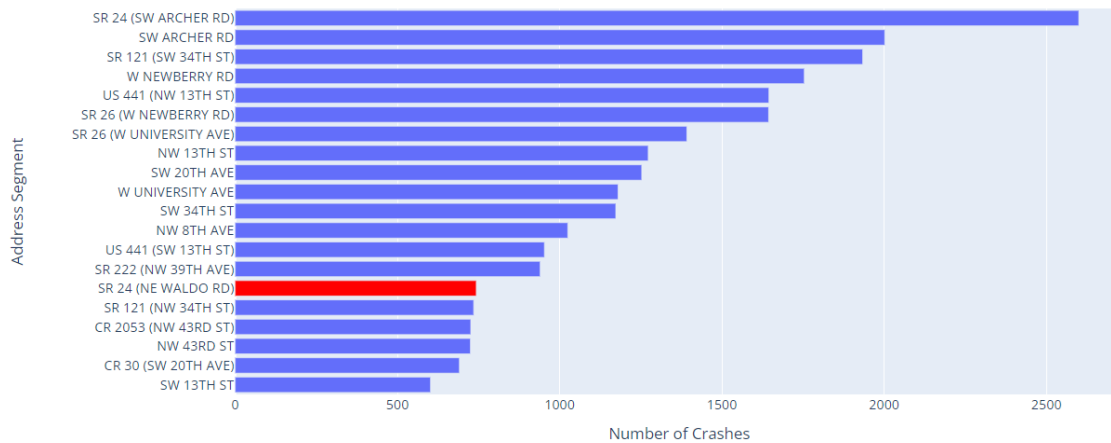
H₀: Crash type and intersection status are not statistically significant predictors of fatality outcomes.

H_a: Crash type and intersection status are statistically significant predictors of fatality outcomes.

The model analysis for crash types and intersection status is statistically significant. Pedestrian involvement is a highly significant predictor ($p < 0.001$) of fatality outcomes, with a positive coefficient of 1.4072 indicating an increase in the log-odds of fatal outcomes. Conversely, motor vehicle only crashes are associated with a significant decrease in fatality outcomes (coefficient = -1.6958, $p < 0.001$). For the intersection type, we included a dummy variable called “AtIntersection,” which encompasses different intersection types (four-way, t/y intersections, etc.). Interestingly, the AtIntersection variable yielded a p-value of 0.331, suggesting that within this specific model, the presence of an intersection is not a statistically significant predictor of fatality.



Top 20 Crash Address Segments (Highlighted by Highest Fatality)



Our project also explored the primary geographic hotspots and peak temporal windows for traffic accidents in Gainesville. The time-based analysis showed that crashes peaked during mornings (8-9 AM) and afternoons/evenings (3-6 PM). A higher proportion of fatal crashes occurs during late night and early morning hours (1-5 AM). The location-based analysis identified recurring high-crash roads like SW Archer Rd and SW 34th St.

Conclusion

These findings have several implications for local government and transportation policy in Gainesville. First, while the city has already made significant investments in traffic safety and infrastructure, including wider sidewalks, additional crosswalks, bike lanes, roundabouts, speed reduction measures, and Vision Zero initiatives, our results suggest that future efforts should place greater emphasis on how these improvements are targeted and refined, particularly for vulnerable users in high-crash corridors and areas with heavy pedestrian, bicycle, and micromobility activity. Rather than expanding infrastructure uniformly, the next phase of investment should focus on locations and crash types associated with higher severity risk. In addition, attention should be given not only to the presence of infrastructure, but also to its clarity and consistency, ensuring that roadway design, signage, and traffic controls are easily understood by all users and do not create confusion in more complex environments such as mixed one-way and multi-directional intersections. Second, because crash activity is concentrated during specific times of day, enforcement and engineering responses may be more effective if they are targeted toward these higher-risk periods rather than applied uniformly across all hours. Finally, Gainesville should continue aligning transportation planning with broader local changes discussed in this paper, including population growth, evolving commuting patterns, and the rise of scooters and e-bikes, while also incorporating ongoing evaluation of existing safety measures so that policies can be adjusted over time based on observed crash outcomes.

At the same time, this study has limitations. As noted in the project draft, the geographic mapping analysis may not fully capture crashes on UF's main campus, and many observations map to the same longitude and latitude despite different listed addresses, which limits the precision of hotspot analysis. Even with these limitations, the analysis still provides a useful picture of long-run crash patterns in Gainesville and offers a stronger basis for thinking about future traffic safety improvements.

Overall, the main takeaway from this study is that Gainesville's traffic safety challenge is not only about reducing the number of crashes, but also about understanding which users face the greatest severity risk and where those risks are most concentrated. The city appears to have made progress in reducing total crashes, especially motor vehicle only crashes, but the patterns involving pedestrians and cyclists show why vulnerable user safety should remain a central priority in future planning and policy decisions. If Gainesville hopes to make meaningful progress toward its Vision Zero goals, success should be measured not only by whether crash totals fall, but by whether the city becomes measurably safer for its most vulnerable road users.

Appendix

- 1 Gainesville, Florida Population 2026. (2026-04-11). World Population Review. <https://worldpopulationreview.com/us-cities/florida/gainesville>
- 2 QuickFacts, Gainesville city, Florida. (2025-07-01). United States Census Bureau. <https://www.census.gov/quickfacts/fact/table/gainesvillecityflorida/>
- 3 The University of Florida At A Glance (2025). UF News. <https://news.ufl.edu/for-media/uf-facts/>
- 4 University Context Area Analysis. (2019-10). Planning, Design & Construction Division, University of Florida. <https://pdc.ufl.edu/wp-content/uploads/nox/pdc/resources/campus-planning/university-context-area/ufcaa-1-UFContextAreaAnalysis2020-2030.pdf>
- 5 Gainesville, FL Census Place. (2024). Data USA. <https://datausa.io/profile/geo/gainesville-fl/>
- 6 "Fuchs: All University of Florida employees to return to in-person work over next 30 days". (2025-07-24). Gainesville Sun. <https://www.gainesville.com/story/news/education/2025/07/24/fuchs-all-uf-employees-to-return-to-in-person-work-over-next-30-days/85358358007/>
- 7 Seth Johnson, "GNV eyes change for micromobility program". (2022-07-29). Main ST Daily News. <https://www.mainstreetdailynews.com/govt-politics/gnv-eyes-change-for-micromobility-program>
- 8 Transportation Improvement Program. (2022-07-11). Metropolitan Transportation Planning Organization for the Gainesville Urbanized Area. <https://ncfrpc.info/mtpo/publications/TIP/2022/TIPDOC22c.pdf>
- 9 City of Gainesville, "Gainesville begins first roadway projects funded by infrastructure surtax". (2023-12-23). Alachua Chronicle. <https://alachuachronicle.com/gainesville-begins-first-roadway-projects-funded-by-infrastructure-surtax/>
- 10 City of Gainesville Vision Zero Action Plan 2024-2028. (2024). Department of Transportation. https://www.gainesvillefl.gov/files/assets/public/v/1/transportation/images/photos-vision-zero/gnvz_actionplan_2024-2028.pdf
- 11 Traffic Crashes Dataset. (2026-04-07). DataGNV – Gainesville's Open Data Portal. https://data.cityofgainesville.org/Public-Safety/Traffic-Crashes/iecn-3sxx/about_data